

Relation Algebra

§14. Thinning Algorithms

§14.1. Thinning

(14.1a)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$ (8.1).

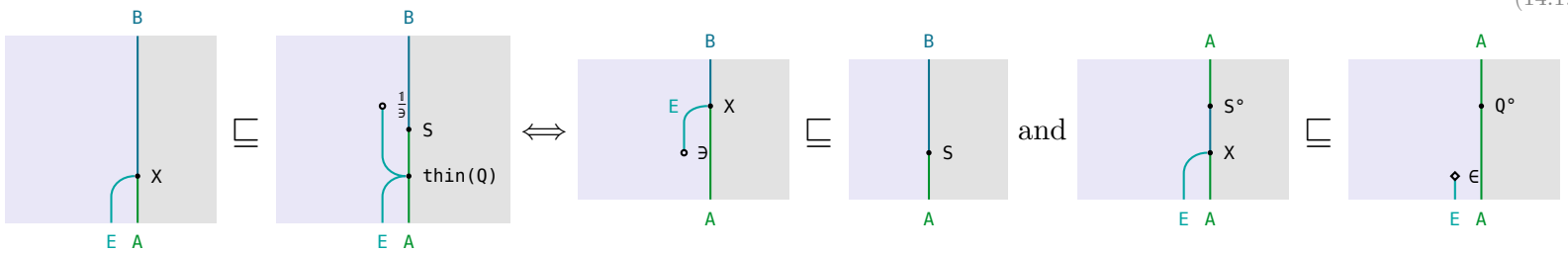
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(14.1b)

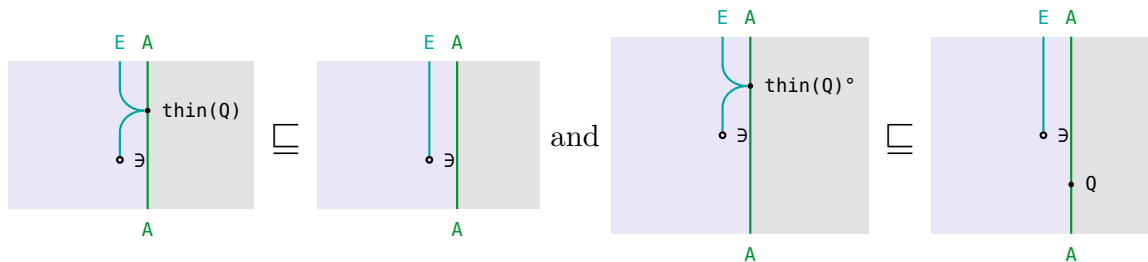
the law	what it says
$X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$	everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
$\text{thin}(Q) \exists \subseteq \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \subseteq \exists Q$	every element of the set has a Q -lower bound among the kept ones
$Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \subseteq Q \implies \text{thin}(Q) \text{thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \subseteq R \wedge 1 \subseteq Q \wedge R R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \text{est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (14.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \subseteq R$ follows	thin-introduction : thinning first cannot lose an R -minimum
$\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists}$ (8.2) — (14.1.2d)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{union} \subseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§14.1.1. $X \subseteq (S^\circ \exists) \text{thin}(Q) \iff X \exists \subseteq S$ and $S^\circ X \subseteq Q^\circ \epsilon$

(14.1.1a)



(14.1.1b)



(14.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \exists = \wedge(\exists) = 1$

§14.1.2. $\text{est}(R) = \text{thin}(Q) \text{est}(R)$ given $Q \subseteq R$, $1 \subseteq Q$, $R R \subseteq R$

$\epsilon \text{ thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$ (14.1.2c)	
	the $\exists$ half, second condition — $Q \sqsubseteq R$, R transitive
$A \text{---} \boxed{\epsilon} \text{---} \boxed{\text{thin}(Q)} \text{---} \epsilon A \text{---} \boxed{\text{est}(R)} \text{---} A$	
$\sqsubseteq$ $A \text{---} \boxed{Q} \text{---} \boxed{\epsilon} \text{---} \epsilon A \text{---} \boxed{\text{est}(R)} \text{---} A$	$\epsilon \text{ thin}(Q) \sqsubseteq Q^\circ \epsilon$ — (14.1.1a) at $S \triangleq \exists, X \triangleq \text{thin}(Q)$, second half
$\sqsubseteq$ $A \text{---} \boxed{Q} \text{---} \boxed{R} \text{---} A$	$\epsilon \text{ est}(R) \sqsubseteq R^\circ$ — est-laws first row at $X \triangleq \text{est}(R)$, second half, conversed
$\sqsubseteq$ $A \text{---} \boxed{R} \text{---} \boxed{R} \text{---} A$	$Q \sqsubseteq R$, conversed
$\sqsubseteq$ $A \text{---} \boxed{R} \text{---} A$	R transitive, conversed

$\text{est}(Q) \frac{1}{3} \sqsubseteq \text{thin}(Q)$ (14.1.2d)	
the singleton holding a Q -least member of a set is a thinning of that set	
circuit the $\exists/\exists$ half of $\triangleq$ thin at $X \triangleq \text{est}(Q)$ $\frac{1}{3}$	Hinze–Marsden
$\frac{1}{3} \exists = 1$ — pow-laws — then $\text{est}(Q) \sqsubseteq \exists$ — est-laws	
the $\epsilon \setminus (Q^\circ \epsilon)$ half of $\triangleq$ thin, $\neg \sqsubseteq Q^\circ \epsilon$	
$\epsilon \text{ est}(Q) \sqsubseteq Q^\circ$ — est-up at $X \triangleq \text{est}(Q)$, conversed	
$\frac{1}{3} \sqsubseteq \epsilon$, since $\frac{1}{3} \exists = 1$ with $\frac{1}{3}$ a map — pow-laws	

$\text{Rn}S^\circ S \sqsubseteq Q \Rightarrow \frac{S}{S} \text{est}(R) \frac{1}{S} \sqsubseteq \frac{S}{S} \text{thin}(Q)$ given $\text{Rn}(S^\circ S) \sqsubseteq Q$, Q a preorder	
circuit	Hinze–Marsden
$\leftarrow B \text{---} \boxed{\frac{1}{S}} \text{---} EB \text{---} \boxed{E(S)} \text{---} EA \text{---} \boxed{\text{est}(R)} \text{---} A \text{---} \boxed{\frac{1}{S}} \text{---} EA \text{---} \boxed{\exists} \text{---} A$ the first of (14.1b)‘ two conditions at $X \stackrel{S}{=} \text{est}(R) \frac{1}{S}$	
$\sqsubseteq B \text{---} \boxed{S} \text{---} A$ $\frac{1}{S} \exists = 1$ — pow-laws — then $\frac{S}{S} \text{est}(R) = S \cap (S^\circ \setminus R^\circ) \sqsubseteq S$ — est-laws	
$\leftarrow A \text{---} \boxed{S} \text{---} B \text{---} \boxed{\frac{1}{S}} \text{---} EB \text{---} \boxed{E(S)} \text{---} EA \text{---} \boxed{\text{est}(R)} \text{---} A \text{---} \boxed{\frac{1}{S}} \text{---} EA$ the second condition, $\neg Q^\circ \in$	
$= A \text{---} \boxed{S} \text{---} B \text{---} \boxed{\frac{1}{S}} \text{---} EB \text{---} \boxed{E(S)} \text{---} EA \text{---} \boxed{\text{est}(\text{Rn}S^\circ S)} \text{---} A \text{---} \boxed{\frac{1}{S}} \text{---} EA$ $\frac{S}{S} \text{est}(R) = \frac{S}{S} \text{est}(\text{Rn}S^\circ S)$ — est-laws	
$\sqsubseteq A \text{---} \boxed{\epsilon} \text{---} EA \text{---} \boxed{\text{est}(\text{Rn}S^\circ S)} \text{---} A \text{---} \boxed{\frac{1}{S}} \text{---} EA$ $S^\circ \frac{S}{S} \sqsubseteq \epsilon$, since $\frac{S}{S} \exists = S$ with $\frac{S}{S}$ a map — pow-laws	
$\sqsubseteq A \text{---} \boxed{Q} \text{---} \boxed{\frac{1}{S}} \text{---} EA$ $\epsilon \text{est}(\text{Rn}S^\circ S) \sqsubseteq (\text{Rn}S^\circ S)^\circ$ — est-up at $X \stackrel{S}{=} \text{est}(\text{Rn}S^\circ S)$, conversed — then $\text{Rn}(S^\circ S) \sqsubseteq Q$	
$\sqsubseteq A \text{---} \boxed{Q} \text{---} \boxed{\epsilon} \text{---} EA$ $\frac{1}{S} \sqsubseteq \epsilon$, since $\frac{1}{S} \exists = 1$ with $\frac{1}{S}$ a map — pow-laws	

$\text{PreservesRecip}(F) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R \sqsubseteq R \wedge \text{MonotonicAlg}(S, Q) \Rightarrow \langle \frac{F(\exists)S}{\exists} \text{thin}(Q) \rangle \text{est}(R) \sqsubseteq \langle \frac{(S)}{\exists} \text{est}(R) \rangle$ (14.1.2g)	
the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \sqsubseteq R$, both preorders	
circuit	Hinze–Marsden
$\sqsubseteq$ thinning theorem	
$=$ $\text{est}(R) = \text{thin}(Q) \text{ est}(R) \quad \text{— (14.1b), } Q \sqsubseteq R$	

§14.2. Paths in a layered network

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.
 $\text{wrapz} \triangleq (\text{wrap}, \text{zero})$, $\text{consw}(a, (xs, n)) = (\text{cons}(a, xs), \text{wt}(a, \text{head}(xs)) + n)$.
 $\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.
 $Q \triangleq R \cap (\text{head} \text{ head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{ est}(R)$.

the law	what it says
$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$\text{PreservesRecip}(F) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ R^\circ \sqsubseteq R^\circ \wedge \text{MonotonicAlg}(F(\exists, 1) \alpha, Q) \wedge R \cap (F(1, \exists) \alpha)^\circ F(1, \exists) \alpha \sqsubseteq Q \Rightarrow \langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R)) \rangle \text{est}(R) \sqsubseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \text{est}(R) \rangle$ (14.2c)	
a least-cost path in a layered network, as a fold over the layers	
circuit	Hinze–Marsden
$T(EA) \rightarrow \left[\frac{1}{\exists} \right] \rightarrow E(T(EA)) \rightarrow E(F(\exists, 1) \alpha) \rightarrow EB \rightarrow \text{est}(R) \rightarrow B$ $= \frac{L(\exists)}{\exists} \text{est}(R)$	
$\sqsubseteq$ (14.1.2g), at $F(\exists, 1) \alpha$ monotonic on Q — (14.2b).	
$\sqsubseteq$ (14.2d) under $\langle \quad \rangle$ monotonic: the whole chain runs inside the reduce, and the $\text{est}(R)$ behind it never moves.	

$\text{Rn}(F(1, \exists)\alpha) \circ F(1, \exists)\alpha \sqsubseteq Q \implies \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R)) \sqsubseteq \frac{F(\exists, \exists)\alpha}{\exists} \text{thin}(Q) \quad (14.2d)$ thinning the algebra of a layered network costs no more than taking the program's two cases	
circuit	Hinze–Marsden
$F(EA)(EB) \xrightarrow{\frac{1}{\exists}} E(F(EA)(EB)) \xrightarrow{E(F(\exists, \exists)\alpha)} \text{thin}(Q) \xrightarrow{} EB$ the algebra of (14.2c) row 2	
$C \xrightarrow{\frac{1}{\exists}} EC \xrightarrow{E(V)} EW \xrightarrow{P(\frac{S}{\exists})} EEA \xrightarrow{\text{union}} EA$ $= F(\exists, \exists) = F(\exists, 1)F(1, \exists); \frac{F(\exists, 1)F(1, \exists)\alpha}{\exists} = \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)\alpha}{\exists}) \text{union}.$	
$\sqsubseteq EEA \xrightarrow{P(\text{thin}(Q))} \text{union} \xrightarrow{} EA$ union thin(Q) $\sqsupseteq$ P(thin(Q)) union — (14.1b).	
$\sqsubseteq B \xrightarrow{\frac{1}{\exists}} EB \xrightarrow{E(S)} EA \xrightarrow{\text{est}(R)} A \xrightarrow{\frac{1}{\exists}} EA$ $\frac{S}{\exists} \text{thin}(Q) \sqsupseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ — (14.1b) at $S \triangleq F(1, \exists)\alpha$, $\text{Rn}(S^\circ S) \sqsubseteq Q$ — (14.2b).	
$= F(EA)(EB) \xrightarrow{\frac{1}{\exists}} E(F(EA)(EB)) \xrightarrow{E(F(\exists, 1))} E(FA)(EB) \xrightarrow{P(\frac{F(1, \exists)\alpha}{\exists} \text{est}(R))} EB$ $P(\frac{1}{\exists}) \text{union} = 1.$	
$= V + V \times \text{pow}(\text{list}^+(V)) \xrightarrow{} \text{list}^+(V)$ α a map, so $\frac{F(1, \exists)\alpha}{\exists} = \frac{F(1, \exists)}{\exists} P(\alpha); \frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] \triangleq \text{cost}.$ $= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangleq \text{cost}$	

§14.3. Implementing thin

(14.3a)

$\text{setify} : [A] \rightarrow EA$, $\text{cup} : EA \times EA \rightarrow EA$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P$ for P a connected preorder.

$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $\langle [\text{nil}, \text{bump } Q] \rangle$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

(14.3b)

the law	what it says
$\text{thinlist}(Q) \text{ xs} = [\text{minlist}(Q) \text{ xs}]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(P) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(P)$ (8.6) — (14.3c)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(P) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f P f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(P)$ (8.8)	shunt a function through a sort
$\text{sort}(P) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(P)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(P) \times \text{sort}(P)) \text{ merge}(P) \sqsubseteq \text{cup sort}(P)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(P)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(FP)$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered subseq $\sqsubseteq$ subseq ordered $\wedge$ thinlist setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sortRel(L, setify, ordered) thinlist $\sqsubseteq$ thin(Q) sortRel(L, setify, ordered) (14.3c)

a thinning of the sorted list lists a thinning of the set — P a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.

circuit	Hinze–Marsden
$EA \text{ --- } \boxed{\text{setify}} \text{ --- } \boxed{\text{ordered}} \text{ --- } \boxed{\text{thinlist}} \text{ --- } LA$	
$\begin{aligned} &EA \text{ --- } \boxed{\text{setify}} \text{ --- } \boxed{\text{ordered}} \text{ --- } \boxed{\text{thinlist}} \text{ --- } LA \\ = &\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ --- } \triangleq \text{thinlist} \end{aligned}$	
$\begin{aligned} &EA \text{ --- } \boxed{\text{setify}} \text{ --- } \boxed{\text{thinlist}} \text{ --- } \boxed{\text{ordered}} \text{ --- } LA \\ \sqsubseteq &\text{ordered } P \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q) \text{ ordered } P, \text{ since } \text{thinlist}(Q) \sqsubseteq \text{subseq} \\ &\text{--- } \triangleq \text{thinlist} \text{ --- and a subsequence of a } P\text{-ordered list is } P\text{-ordered} \end{aligned}$	
$\begin{aligned} &EA \text{ --- } \boxed{\text{thin}(Q)} \text{ --- } \boxed{\text{setify}} \text{ --- } \boxed{\text{ordered}} \text{ --- } LA \\ \sqsubseteq &\triangleq \text{thinlist's thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ after setify}^\circ, \text{ at} \\ &\text{setify}^\circ \text{ setify} \sqsubseteq 1 \text{ for setify simple --- dom-laws --- then } \cdot \text{setify} \cdot \text{setify}^\circ \text{ --- triple-chains} \end{aligned}$	
$\begin{aligned} &EA \text{ --- } \boxed{\text{thin}(Q)} \text{ --- } \boxed{\text{setify}} \text{ --- } \boxed{\text{ordered}} \text{ --- } LA \\ = &\text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ --- } \triangleq \text{thinlist} \end{aligned}$	

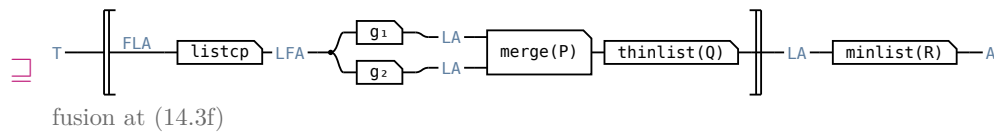
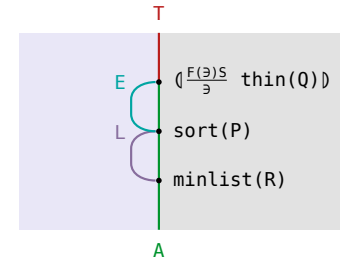
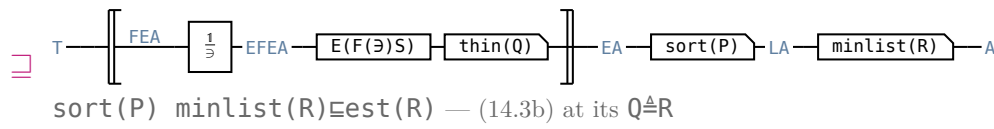
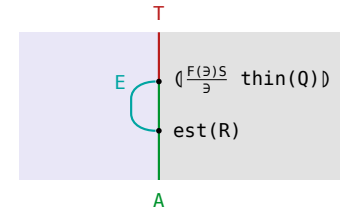
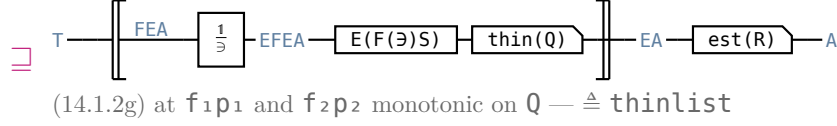
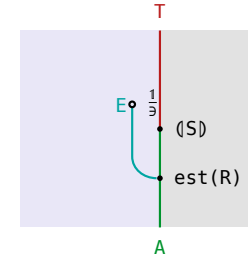
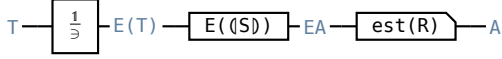
$\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sort}F(X) \sqsubseteq \text{sort}F(Y)) \wedge \text{MonotonicAlg}(f, P) \wedge$ $\text{sort}F(fP f^\circ) \text{list}f \sqsubseteq P(f) \text{sort}(P) \wedge \text{sort}(P) \text{filter}p \sqsubseteq E(p) \text{sort}(P) \wedge$ $F(\text{sort}(P)) \text{list}cp \sqsubseteq cp(F, A) \text{sort}F(F(P)) \Rightarrow F(\text{sort}(P)) \text{list}cp \text{list}f \text{filter}p \sqsubseteq \frac{F(\exists)f p}{\exists} \text{sort}(P)$ one sorted list built from sorted arguments, instead of a set built and then sorted — $f : FA \rightarrow A$ monotonic on P, p coreflexive, F linear.		(14.3d)
circuit	Hinze–Marsden	
$FEA \rightarrow \boxed{\frac{1}{\exists}} \rightarrow EFA \rightarrow \boxed{E(F(\exists)f p)} \rightarrow EA \rightarrow \boxed{\text{sort}(P)} \rightarrow LA$		
$= FEA \rightarrow \boxed{cp(F, A)} \rightarrow EFA \rightarrow \boxed{E(fp)} \rightarrow EA \rightarrow \boxed{\text{sort}(P)} \rightarrow LA$ $= \frac{F(\exists)f p}{\exists} = \frac{F(\exists)}{\exists} E(fp) \text{ — pow-laws; } cp(F) \triangleq \frac{F(\exists)}{\exists} \text{ — } \triangleq \text{thinlist}$		
$= FEA \rightarrow \boxed{cp(F, A)} \rightarrow EFA \rightarrow \boxed{P(f)} \rightarrow \boxed{E(p)} \rightarrow EA \rightarrow \boxed{\text{sort}(P)} \rightarrow LA$ $= E(fp) = E(f)E(p); E(f) = P(f) \text{ for } f \text{ a map — powrel-laws}$		
$\sqsubseteq FEA \rightarrow \boxed{cp(F, A)} \rightarrow EFA \rightarrow \boxed{P(f)} \rightarrow EA \rightarrow \boxed{\text{sort}(P)} \rightarrow \boxed{\text{filter}p} \rightarrow LA$ $\text{sort}(P) \text{filter}(p) \sqsubseteq E(p) \text{sort}(P) \text{ — (14.3b)}$		
$\sqsubseteq FEA \rightarrow \boxed{cp(F, A)} \rightarrow EFA \rightarrow \boxed{\text{sort}F(fP f^\circ)} \rightarrow LFA \rightarrow \boxed{\text{list}f} \rightarrow \boxed{\text{filter}p} \rightarrow LA$ $\text{sort}(fP f^\circ) \text{list}(f) \sqsubseteq P(f) \text{sort}(P) \text{ — (14.3b)}$		
$\sqsubseteq FEA \rightarrow \boxed{cp(F, A)} \rightarrow EFA \rightarrow \boxed{\text{sort}F(F(P))} \rightarrow LFA \rightarrow \boxed{\text{list}f} \rightarrow \boxed{\text{filter}p} \rightarrow LA$ $FP \sqsubseteq fP f^\circ \text{ — mon-str at } f \text{ a map; } \text{sort}(P) \triangleq \text{setify}^\circ \text{ ordered } P \text{ grows with } P \text{ — } \triangleq \text{thinlist}$		
$\sqsubseteq FEA \rightarrow \boxed{F(\text{sort}(P))} \rightarrow FLA \rightarrow \boxed{\text{list}cp} \rightarrow LFA \rightarrow \boxed{\text{list}f} \rightarrow \boxed{\text{filter}p} \rightarrow LA$ $F(\text{sort}(P)) \text{list}cp \sqsubseteq cp(F) \text{sort}(FP) \text{ — (14.3b), } F \text{ linear}$		

$\text{PreservesRecip}(F) \wedge \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \in R \wedge 1 \in Q \wedge Q \subseteq Q \wedge R^\circ R^\circ \subseteq R^\circ \wedge \text{MonotonicAlg}(f_1 \ p_1, Q) \wedge$ (14.3e)
 $\text{MonotonicAlg}(f_2 \ p_2, Q) \wedge (\forall X. \forall Y. X \in Y \implies \text{sort}F(X) \in \text{sort}F(Y)) \wedge \text{MonotonicAlg}(f_1, P) \wedge$
 $\text{MonotonicAlg}(f_2, P) \wedge \text{sort}F(f_1 \ P f_1^\circ) \text{list}f_1 \in P(f_1) \text{sort}(P) \wedge \text{sort}F(f_2 \ P f_2^\circ) \text{list}f_2 \in P(f_2) \text{sort}(P) \wedge$
 $\text{sort}(P) \text{filter}p_1 \in E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filter}p_2 \in E(p_2) \text{sort}(P) \wedge$
 $F(\text{sort}(P)) \text{list}cp \in cp(F, A) \text{sort}F(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \in \text{cup} \text{sort}(P) \wedge$
 $\text{sort}(P) \text{thinlist}(Q) \in \text{thin}(Q) \text{sort}(P) \wedge \text{sort}(P) \text{minlist}(R) \in \text{est}(R) \wedge g_1 = \text{list}f_1 \text{ filter}p_1 \wedge$
 $g_2 = \text{list}f_2 \text{ filter}p_2 \wedge S = f_1 \ p_1 \cup f_2 \ p_2 \implies (\text{list}cp(g_1, g_2) \text{merge}(P) \text{thinlist}(Q)) \text{minlist}(R) \in \frac{(S)}{3}$
 $\text{est}(R)$

a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

circuit

Hinze–Marsden



$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \in Y \Rightarrow \text{sortF}(X) \in \text{sortF}(Y)) \wedge \text{MonotonicAlg}(f_1, P) \wedge \\
& \text{MonotonicAlg}(f_2, P) \wedge \text{sortF}(f_1 \text{ Pf}_1^\circ) \text{listf}_1 \in P(f_1) \text{sort}(P) \wedge \text{sortF}(f_2 \text{ Pf}_2^\circ) \text{listf}_2 \in P(f_2) \text{sort}(P) \wedge \\
& \text{sort}(P) \text{filterp}_1 \in E(p_1) \text{sort}(P) \wedge \text{sort}(P) \text{filterp}_2 \in E(p_2) \text{sort}(P) \wedge \\
& F(\text{sort}(P)) \text{listcp} \in \text{cp}(F, A) \text{sortF}(F(P)) \wedge (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \in \text{cup} \text{sort}(P) \wedge \\
& \text{sort}(P) \text{thinlist}(Q) \in \text{thin}(Q) \text{sort}(P) \Rightarrow F(\text{sort}(P)) \text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \\
& \text{merge}(P) \text{thinlist}(Q) \in \frac{F(\exists)(f_1 \text{ p}_1 \cup f_2 \text{ p}_2)}{\exists} \text{thin}(Q) \text{sort}(P)
\end{aligned} \tag{14.3f}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit	Hinze–Marsden
$\sqsubseteq$ $\text{sort}(P) \text{thinlist}(Q) \in \text{thin}(Q) \text{sort}(P) \text{ — (14.3b)}$	
$=$ $\{ \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \} \text{cup} \text{sort}(P) \text{thinlist}(Q)$ $S = (f_1 p_1) \cup (f_2 p_2) \text{ — } \triangleq \text{thinlist, then } \triangleq \text{cup}$	
$\sqsubseteq$ $\{ \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \} (\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \text{thinlist}(Q)$ $(\text{sort}(P) \times \text{sort}(P)) \text{merge}(P) \in \text{cup} \text{sort}(P) \text{ — (14.3b)}$	
$=$ $\{ \frac{F(\exists)f_1 p_1}{\exists} \text{sort}(P), \frac{F(\exists)f_2 p_2}{\exists} \text{sort}(P) \} \text{merge}(P) \text{thinlist}(Q)$ $\langle X, Y \rangle (\text{sort}(P) \times \text{sort}(P)) = \langle X \text{sort}(P), Y \text{sort}(P) \rangle \text{ — bdm-prod-laws}$	
$\sqsubseteq F(\text{sort}(P)) \text{listcp} \langle g_1, g_2 \rangle \text{merge}(P) \text{thinlist}(Q)$ $(14.3d) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X \langle g_1, g_2 \rangle \in \langle Xg_1, Xg_2 \rangle \text{ — bdm-prod-laws; } g_i \triangleq \text{list}(f_i) \text{filter}(p_i) \text{ — } \triangleq \text{thinlist}$	

§14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}, \text{value} \triangleq \text{list}(\text{vol}) \text{ sum}, \text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}.$

$\text{subseq} = ([\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2]) \circ \text{w}$, within w the coreflexive on xs with weight $xs \leq w$, $0 \leq w$.

$R \triangleq \text{value} \geq \text{value}^\circ$, $Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ)$, $P \triangleq R \cup Q$.

$FA = 1 + \text{Item} \times A$, $\text{listcp} = \text{wrap} + \text{cpr}$, $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])$ $\text{filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$,
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2].$

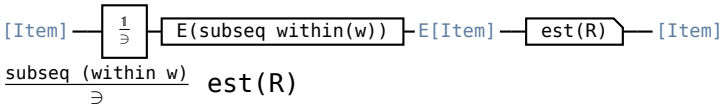
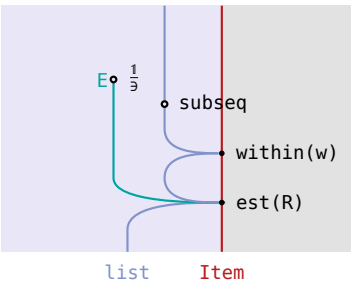
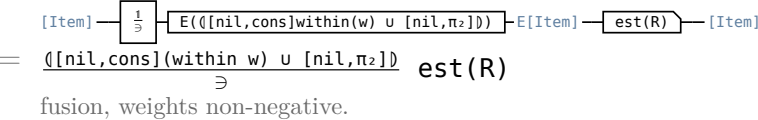
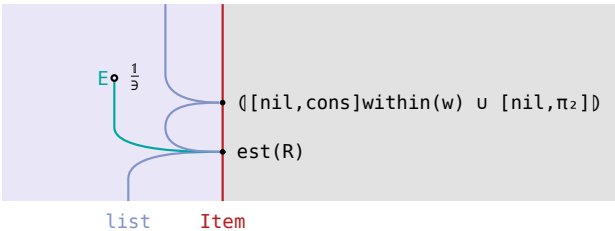
$h_1 \triangleq \text{list}(\text{cons}) \text{filter}(\text{within } w)$, $h_2 \triangleq \text{list}(\pi_2).$

(14.4b)

the law	what it says
$(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
$(1 \times Q) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$$\begin{aligned}
 & 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\
 & \text{sortF}([\text{nil}, \text{cons}]R[\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sortP} \wedge \\
 & \text{sortF}([\text{nil}, \pi_2]R[\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \text{sortP} \wedge \text{sortP } \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sortP} \wedge \\
 & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, [\text{Item}]) \text{sortF}(F(R)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup } \text{sortP} \wedge \text{sortP} \\
 & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sortP} \wedge \text{sortP } \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp}(\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \\
 & \text{mergeP } \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{subseq } \text{within}(w)}{\supset} \text{est}(R)
 \end{aligned}
 \tag{14.4c}$$

the knapsack problem, as a fold that thins the packings kept at each item

circuit	Hinze–Marsden
	
	
 $\sqsubseteq (\text{listcp } (g_1, g_2) \text{ merge } R \text{ thinlist}(Q))$ $\text{minlist}(R)$ binary thinning theorem, at $P \triangleq R$, F linear, Q from (14.4b)	
$([\text{nil}, \text{cpr } (h_1, h_2) \text{ merge } R \text{ thinlist}(Q)])$ $\text{minlist}(R)$ $\text{listcp} = \text{wrap} + \text{cpr}$, $g_i = [\text{list}(\text{nil}), h_i]$ — $\triangleq$ within; $\text{minlist}(R)$ is head, packings coming out in descending value	

§14.5. The paragraph problem

(14.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(14.5b)

the law	what it says
(1×R) glue⊆glue R FALSE	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
(1×Q) new⊆new Q (1×Q) (glue (ok w))⊆glue (ok w)Q cons monotonic on collect≤collect°.	both halves are monotonic on Q once ties in waste are broken by the first line
merge T=cat; P≐head prefix head° also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

$$\begin{aligned}
& (\forall a. 0 \leq \text{len}(a)) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap} \\
& \text{wrap}, \text{new}] \tau [\text{wrap} \text{ wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{new}]) \text{sortP} \wedge \text{sortF}([\text{wrap} \text{ wrap}, \text{glue}] \tau [\text{wrap} \\
& \text{wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq \text{P}([\text{wrap} \text{ wrap}, \text{glue}]) \text{sortP} \wedge \text{sortP} \text{filterp}_2 \sqsubseteq \text{E}(\text{ok}(w)) \text{sortP} \wedge \\
& \text{F}(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(\text{F}, \text{list}^+((\text{list}^+ \text{Word}))) \text{sortF}(\text{F}(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{mergeP} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \\
& \text{thinlist} \sqsubseteq \text{thin}(\text{Q}(\text{len}, w)) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(\text{R}) \Rightarrow (\text{listcp}(\text{listf}_1, \text{listf}_2 \text{filterp}_2) \\
& \text{mergeP} \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}(w)}{\supset} \text{est}(\text{R})
\end{aligned} \tag{14.5c}$$

a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}(\text{partition fits}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & \frac{\text{partition list}^+(\text{fits } w)}{\supset} \text{est}(\text{R}) \end{aligned} $	
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))}{\supset} \text{est}(\text{R}) \end{aligned} $ fusion, every word fits on a line by itself.	
$ \begin{aligned} & \text{list}^+(\text{Word}) \xrightarrow{\frac{1}{3}} \text{E}(\text{list}^+(\text{Word})) \xrightarrow{\text{E}([\text{wrap wrap}, \text{new}] \cup [\text{wrap wrap}, \text{glue}] \text{ok}(w))} \text{E}(\text{list}^+(\text{list}^+(\text{Word}))) \xrightarrow{\text{est}(\text{R})} \text{list}^+(\text{list}^+(\text{Word})) \\ & = \frac{([\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] \text{ok}(w)))}{\supset} \text{est}(\text{R}) \end{aligned} $ the algebra as $(f_1 p_1) \cup (f_2 p_2)$, $p_1 \triangleq 1 \text{ — } \triangleq \text{thinlist}$.	
$ \begin{aligned} & \text{listcp}(\text{g}_1, \text{g}_2) \text{cat thinlist}(\text{Q}) \\ & \sqsubseteq \text{minlist}(\text{R}) \end{aligned} $ binary thinning theorem, at $\text{P} \triangleq \text{T}$ with merge $\tau = \text{cat}$, Q from (14.5b)	
$ \begin{aligned} & ([\text{start}, \text{cpr}(\text{h}_1, \text{h}_2) \text{cat thinlist}(\text{Q})]) \\ & = \text{minlist}(\text{R}) \\ & \text{listcp} = \text{wrap} + \text{cpr}, \text{g}_i \text{ along the coproduct — } \triangleq \text{partition} \end{aligned} $	

§14.6. Bitonic tours

$\text{FA} = (\text{City} \times \text{City}) + (\text{City} \times \text{A})$, the base functor of cons-lists of length at least two; $\text{listcp} = \text{wrap} + \text{cpr}$; $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$, neither positive nor symmetric.

$\text{start}(a, b) = ([a, b], [a, b])$, $\text{dropl}(a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{dropr}(a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $\text{tour} \triangleq ([\text{start}, \text{dropl} \cup \text{dropr}])$.

$\text{cost}(xs, ys) = \text{outcost } xs + \text{incost } ys$, $\text{outcost}[a_0, \dots, a_n] = \text{tc}(a_0, a_1) + \dots + \text{tc}(a_{n-1}, a_n)$, $\text{incost}[a_0, \dots, a_n] = \text{tc}(a_1, a_0) + \dots + \text{tc}(a_n, a_{n-1})$.

$\text{next} \triangleq \text{tail head}$, $\text{next2} \triangleq \text{next} \times \text{next}$, $\text{head2} \triangleq \text{head} \times \text{head}$, $\text{R} \triangleq \text{cost} \leq \text{cost}^\circ$, $\text{Q} \triangleq \text{Rn}(\text{next2 next2}^\circ) \text{n}(\text{head2 head2}^\circ)$, $\text{P} \triangleq \text{T}$, $\text{g}_1 \triangleq \text{list}([\text{start}, \text{dropl}])$, $\text{g}_2 \triangleq \text{list}([\text{start}, \text{dropr}])$.

the law	what it says
$(1 \times \text{R}) \text{dropl} \sqsubseteq \text{dropl } \text{R}$ FALSE $(1 \times \text{R}) \text{dropr} \sqsubseteq \text{dropr } \text{R}$ FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
$(1 \times \text{Q}) \text{dropl} \sqsubseteq \text{dropl } \text{Q}$ $(1 \times \text{Q}) \text{dropr} \sqsubseteq \text{dropr } \text{Q}$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input

$ \begin{aligned} & (\forall X. \forall Y. X \sqsubseteq Y \implies \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \\ & \text{sortF}([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq P([\text{start}, \text{droplFn}]) \text{sortP} \wedge \\ & \text{sortF}([\text{start}, \text{droprFn}] \sqcup [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq P([\text{start}, \text{droprFn}]) \text{sortP} \wedge \\ & F(\text{sortP}) \text{listcp} \sqsubseteq \text{cp}(F, \text{dTour}(\text{City})) \text{sortF}(F(\tau)) \wedge (\text{sortP} \times \text{sortP}) \text{cat} \sqsubseteq \text{cup} \text{sortP} \wedge \text{sortP} \\ & \text{thinlist} \sqsubseteq \text{thin}(Q(\text{tc})) \text{sortP} \wedge \text{sortP} \text{minlist} \sqsubseteq \text{est}(R) \implies (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}) \text{minlist} \sqsubseteq \\ & \quad \underbrace{(\text{tourAlg})}_{\exists} \text{est}(R) \\ & \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city} \end{aligned} $ (14.6c)	
circuit	Hinze–Marsden
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \text{tour} \xrightarrow{\exists} \text{est}(R) \end{aligned} $	
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & = \underbrace{([\text{start}, \text{dropl} \sqcup \text{dropr}])}_{\exists} \text{est}(R) \\ & \text{tour} \triangleq ([\text{start}, \text{dropl} \sqcup \text{dropr}]) \triangleq \text{tour} \end{aligned} $	
$ \begin{aligned} & \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} E(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \sqcup [\text{start}, \text{droprFn}])} E(\text{Journey}(\text{City})^2) \xrightarrow{\text{est}(R)} \text{Journey}(\text{City}) \\ & \sqsubseteq (\text{listcp}(g_1, g_2) \text{cat} \text{thinlist}(Q)) \text{minlist}(R) \\ & \text{binary thinning theorem, at } P \triangleq T \text{ with merge } T = \text{cat}, Q \text{ from (14.6b)} \end{aligned} $	
$ \begin{aligned} & ([\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\ & \text{thinlist}(Q)] \text{minlist}(R) \\ & = \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\ & \text{quadratic, two tours added per step} \end{aligned} $	